

Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®

JAIN COLLEGE OF ENGINEERING, BELAGAVI
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Department of Electronics and Communication Engineering

Assignment Report for
Embedded System Design (BEC601)

CERTIFICATE

This is to certify that the assignment project entitled “**Real time multitasking system using FreeRTOS and ESP32**” presented by **Siddharth K. (2JI23EC140), Trupti S. (2JI23EC165), Shyambhavi N. (2JI23EC137) and Vikas G. (2JI23EC173)** are bonafide students of **Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi**, in partial fulfilment for the award of **Bachelor of Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**. It is certified that all corrections/suggestions indicated for assignment project have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of assignment project prescribed for the Bachelor of Engineering degree.

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INTRODUCTION

This project implements a real-time multitasking embedded system using FreeRTOS on the ESP32 microcontroller. The system is developed using Embedded C++ in the Arduino environment and integrates multiple components such as a DHT11 temperature and humidity sensor, LEDs, push buttons, an OLED display, and a buzzer to create a monitoring and alert system.

The system is divided into multiple tasks using FreeRTOS, allowing operations like sensor data acquisition, user input handling, output control, and display updates to run concurrently. Inter-task communication is achieved using semaphores and queues, ensuring efficient and synchronized operation.

Based on temperature readings, the system automatically changes its state to Normal, Warning, or Alert, which is indicated through LEDs and buzzer signals. The OLED display shows real-time system information, while the Serial Monitor provides continuous output for debugging and monitoring.

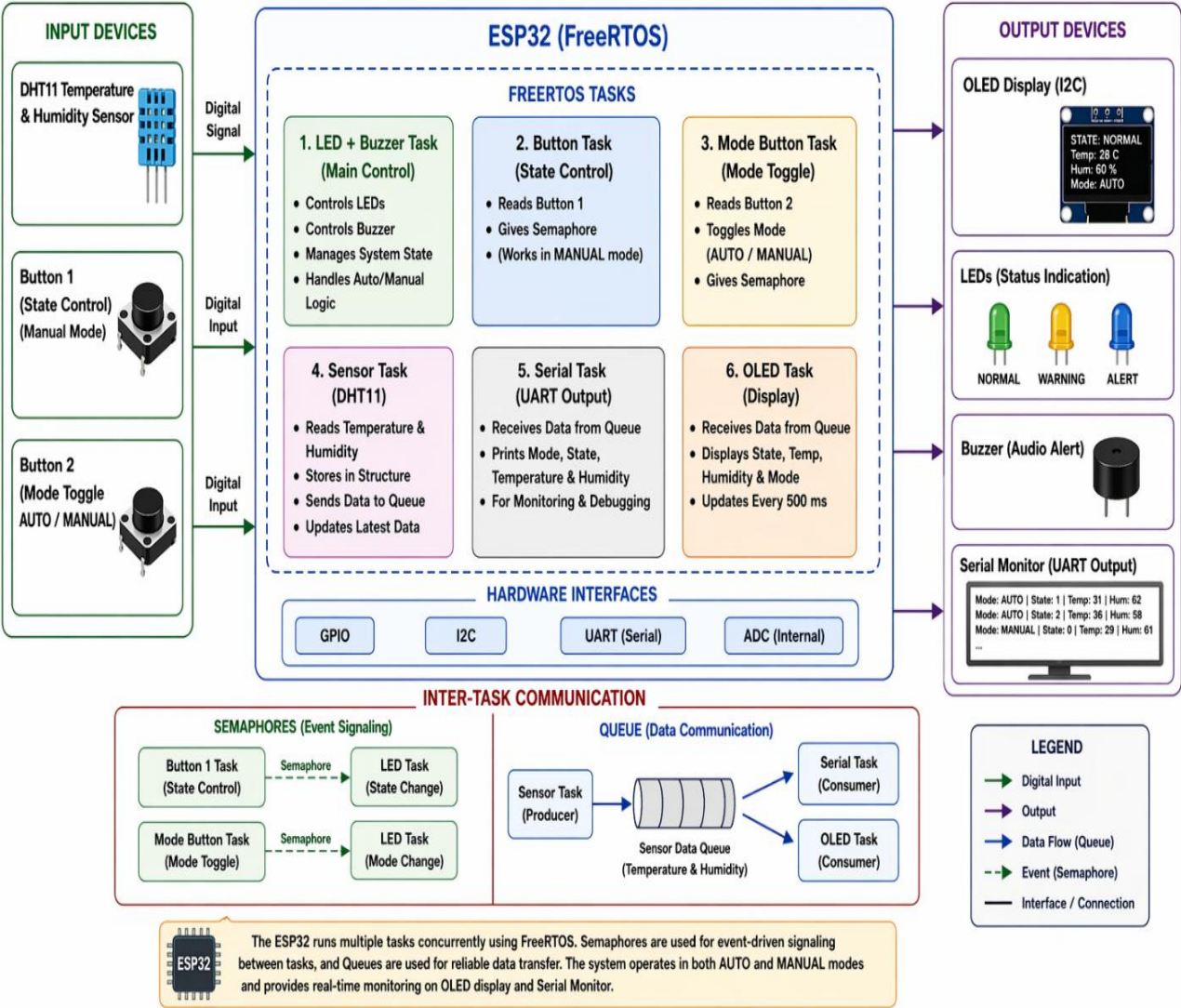
The system operates in two modes: Automatic mode, where decisions are made based on sensor data, and Manual mode, where the user can control the system state using push buttons. A separate button allows switching between these modes.

Overall, this project demonstrates the practical implementation of real-time multitasking, efficient resource management, and integration of multiple peripherals in an embedded system.

SYSTEM ARCHITECTURE

SYSTEM ARCHITECTURE / BLOCK DIAGRAM

Real-Time Multitasking Embedded System using FreeRTOS on ESP32



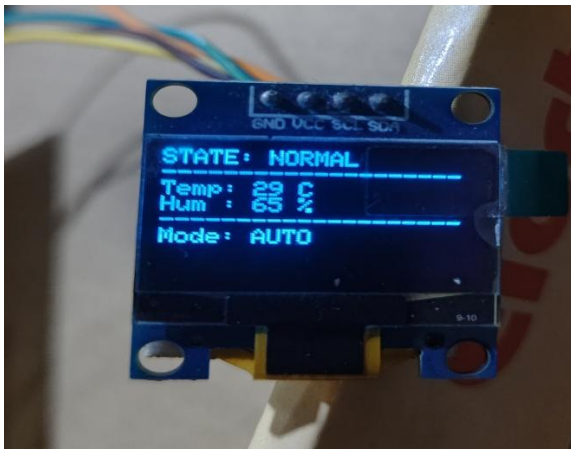
RESULTS AND OUTPUT

Observed Outputs

- **LED Indication:**
 - Green LED → Normal State
 - Yellow LED → Warning State (blinking)
 - Blue LED → Alert State (fast blinking)
- **Buzzer Indication:**
 - OFF → Normal
 - Slow Beep → Warning
 - Fast Beep → Alert
- **OLED Display:**

Displays current state, temperature, humidity, and mode (AUTO/MANUAL) in real-time.
- **Serial Monitor Output:**

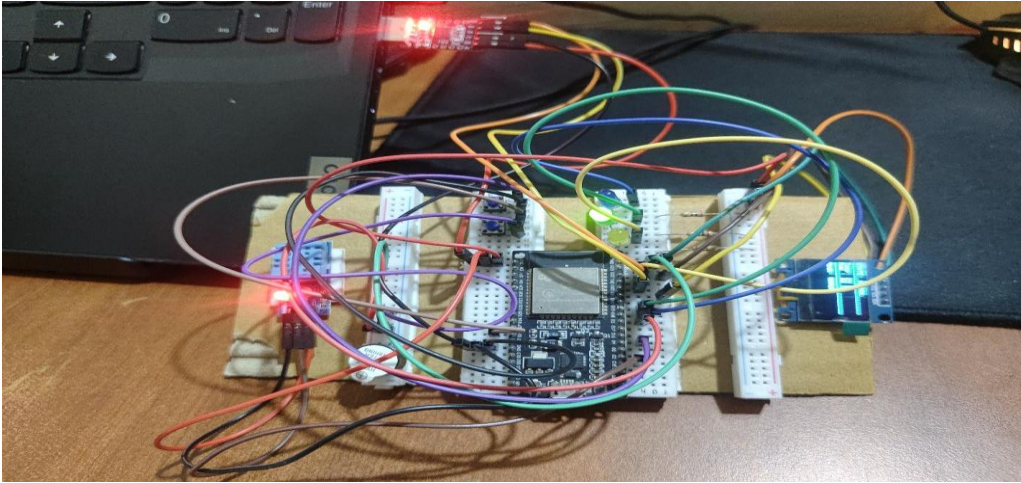
Continuously prints system status including mode, state, temperature and humidity for monitoring and debugging.



OLED OUTPUT

```
Mode: AUTO | State: 0 | Temp: 29 | Hum: 67
Mode: AUTO | State: 0 | Temp: 29 | Hum: 68
Mode: AUTO | State: 0 | Temp: 29 | Hum: 68
Mode: AUTO | State: 0 | Temp: 29 | Hum: 68
Mode: AUTO | State: 0 | Temp: 29 | Hum: 68
Mode: MANUAL | State: 0 | Temp: 29 | Hum: 68
Mode: MANUAL | State: 0 | Temp: 29 | Hum: 68
Mode: MANUAL | State: 0 | Temp: 29 | Hum: 68
Mode: MANUAL | State: 1 | Temp: 29 | Hum: 68
Mode: MANUAL | State: 1 | Temp: 29 | Hum: 67
Mode: MANUAL | State: 2 | Temp: 29 | Hum: 67
Mode: MANUAL | State: 2 | Temp: 29 | Hum: 67
Mode: MANUAL | State: 0 | Temp: 29 | Hum: 68
Mode: AUTO | State: 0 | Temp: 29 | Hum: 68
Mode: AUTO | State: 0 | Temp: 29 | Hum: 68
Mode: MANUAL | State: 0 | Temp: 29 | Hum: 68
```

SERIAL OUTPUT



SYSTEM HARDWARE PROTOTYPE

CODE

Libraries: Arduino.h, DHT.h, Wire.h, Adafruit_GFX.h,
Adafruit_SSD1306.h.

Tasks: void ledTask(void *pvParameters), void buttonTask(void *pvParameters), void modeButtonTask(void *pvParameters), void sensorTask(void *pvParameters), void serialTask(void *pvParameters), void oledTask(void *pvParameters).

FreeRTOS tools: xTaskCreate(), xSemaphoreCreateBinary(),
xSemaphoreTake(), xSemaphoreGive(), xQueueCreate(),
xQueueSend(), xQueueReceive(), vTaskDelay().

CONCLUSION

This project successfully demonstrates a real-time multitasking embedded system using FreeRTOS on the ESP32. Multiple tasks were executed concurrently with efficient communication using semaphores and queues. The system effectively integrates sensing, processing, and output, and supports both automatic and manual modes, making it flexible and reliable.